



Roll No:

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|

**MCA-INT**  
**(SEM III) THEORY EXAMINATION 2024-25**  
**APPLIED LINEAR ALGEBRA**

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

**SECTION A**

1. Attempt all questions in brief.

2 x 7 = 14

|    |                                                                                                                                                                                                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Find: i) $u \cdot v$ and ii) $v \cdot u$ where $u = (1 - 2i, 3 + i)$ and $v = (4 + 2i, 5 - 6i)$ .                                                                                                  |
| b. | Given that $z = 2 + 3i$ and $w = 5 - 2i$ . Find i) $z \cdot w$ , ii) $ z $                                                                                                                         |
| c. | If $T: U \rightarrow V$ is linear transformation from the vector space $U(F)$ to the vector space $V(F)$ , then prove that $T(\alpha - \beta) = T(\alpha) - T(\beta), \forall \alpha, \beta \in U$ |
| d. | Define the Distance and Angle between the nonzero vectors.                                                                                                                                         |
| e. | Find $\ u\ ^2$ , where $u = (1, -2, -4, 5, -3)$ .                                                                                                                                                  |
| f. | What do you mean by Orthogonal Vectors?                                                                                                                                                            |
| g. | Write down the properties of Linear Functional.                                                                                                                                                    |

**SECTION B**

2. Attempt any three of the following:

7 x 3 = 21

|    |                                                                                                                                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Define Linear Span. Determine the subspace spanned by $S = \{(1, 2, 3), (3, 1, 0)\}$ in $R^3$ .                                                                                                                                            |
| b. | Prove that "The relation of isomorphism in any set of vector spaces over a field $F$ is an equivalence relation."                                                                                                                          |
| c. | Find the inverse linear transformation $T$ on $R^2$ defined by $T(x, y) = (y, 2x - y)$ .                                                                                                                                                   |
| d. | State and prove Triangle inequality in an inner product space.                                                                                                                                                                             |
| e. | If $V$ be a finite-dimensional vector over the field $F$ . For each vector $\alpha$ in $V$ define $L_\alpha(f) = f(\alpha)$ for any $f \in V^*$ .<br>The mapping $\alpha \rightarrow L_\alpha$ is then an isomorphism of $V$ onto $V^{**}$ |

**SECTION C**

3. Attempt any one part of the following:

7 x 1 = 7

|    |                                                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Prove that: the Modulo 4 system given by: $(G, +_4, \times_4)$ , where $G = \{0, 1, 2, 3\}$ , is a Ring.                                                                   |
| b. | Define the Vector Space with its axioms.<br>Express $(1, -2, 5)$ as a Linear Combination of the Vectors $\alpha_1 = (1, 0, 1), \alpha_2 = (0, 2, 2), \alpha_3 = (3, 7, 1)$ |

4. Attempt any one part of the following:

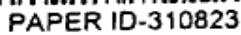
7 x 1 = 7

|    |                                                                                                                                                                   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Show the following mappings are linear: $T: R^2 \rightarrow R$ defined by $T(x, y) = (3x + 5y)$ is a linear Transformation of the vector space.                   |
| b. | Find the matrix of the linear transformation $T$ on $V_3(R)$ defined as: $T(x, y) = \{-x + 2y, y, -3x + 3y\}$ with respect to the basis $B = \{(1, 2), (-2, 1)\}$ |

5. Attempt any one part of the following:

7 x 1 = 7

|    |                                                                                                                                        |
|----|----------------------------------------------------------------------------------------------------------------------------------------|
| a. | Consider that: $\dim V = m$ and $\dim U = n$ .<br>Then Prove that: $\dim[\text{Hom}(V, U)] = mn$ State and proof Rank-nullity Theorem. |
| b. | Consider the Linear Operator $T$ on $R^3$ define by $T(x, y, z) = (2x, 4x - y, 2x + 3y - z)$ . Find i) $T^{-1}$ , ii) $T^2$ .          |

**TIME: 3 HRS**

**M.MARKS: 70**

- $$7 \times 1 = 7$$

$$7 \times 1 = 7$$

<https://www.aktuonline.com>